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## Optimising Windows for Audio

25 days ago · Updated

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This article covers optimisations in the versions of Windows we currently support, Windows 10 and Windows 11. For steps relevant to earlier versions of Windows, please see [Optimising Windows 7 and 8 for audio](#).

Applies to all audio interfaces on Windows.

production.

You might view this article if you experience these problems:

- Pops, clicks, humming and distorted sounds when you record or playback audio.
- A delay between playing your instrument, speaking or singing into a microphone, and hearing the sound from your speakers and headphones – known as latency.
- Your Windows machine runs slowly or keeps crashing.

We do recommend you complete every possible step, even if they seem irrelevant to any issues you are having.



If you're having USB connection or power issues, try this article first:  
[Troubleshooting USB audio interface issues.](#)

## Basic Tips

### Optimise Focusrite software and buffer size

Ensure you have the latest version of the Focusrite driver or control software installed. You can find the latest version for your interface on our [Downloads](#) page: [Focusrite Downloads](#).

Additionally, adjusting buffer size is one of the first things to try to improve audio performance.

Here is how to increase the buffer size if you are experiencing glitchy audio:

1. Right-click on  **Focusrite Notifier** on the taskbar and select your device's settings.

*If you can't see the Focusrite notifier on your taskbar, follow the steps in this article: [The Focusrite Notifier icon isn't on the Windows taskbar – Focusrite](#)*

2. Increase the **Buffer Size** to **1024**.
3. If the performance improves, you can try a lower setting.

*If you are unsure what buffer size is and how it affects performance, please refer to this article:*

[Sample Rate, Bit Depth & Buffer Size Explained](#)

## Check your USB connection

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### Use a suitable USB cable for your hardware

Make sure you use a short USB cable, ideally the one in the box with your interface, or a cable shorter than 2 metres or 6 feet that meets this spec: [Can I use any USB cable with my Focusrite USB Interface?](#)

Using longer USB cables with your interface can cause dropouts, glitches, or prevent your interface from working.

**Make sure any USB cables you use meet the spec in this article:**

[Can I use any USB cable with my Focusrite USB Interface?](#)

### Ensure your interface has its own dedicated USB port

USB hubs are known to cause issues with audio devices. These issues can include noise, and connection problems. Therefore, avoid using them and connect your interface directly to your

## Avoid USB adaptors

USB adaptors are also known to cause issues with audio devices. Therefore, we recommend using a cable with connectors that can connect directly to your interface and your computer's USB ports. If your audio interface comes with an adaptor in the box, this has been tested to ensure it will not cause the same issues that other adaptors might.

*For more information regarding USB cables, hubs and adaptors, please refer to this article: [Can I use any USB cable with my Focusrite USB Interface?](#)*

## Thunderbolt only ⚡ – Check your Thunderbolt connection



Many Focusrite interfaces have Type-C connectors. While Thunderbolt 3 and above do use Type-C connectors, your interface won't be Thunderbolt unless you see the ⚡ **Thunderbolt** symbol next to the Type-C port.

If you are using a Thunderbolt interface, such as Clarett Thunderbolt, please refer to the Clarett Thunderbolt troubleshooting steps. These cover extra steps that are relevant to Thunderbolt interfaces: [Clarett Thunderbolt Windows | Troubleshooting Guide](#).

## Close all other background applications

When you run your DAW (Digital Audio Workstation), make sure you close applications you are not using, like these:

- Music streaming apps (Spotify etc.)
- Web browsers (Chrome, Edge etc.)
- Office applications (Word, Excel etc.)

This will prevent background apps slowing your system down, allowing it to focus on the apps you need.

updating the BIOS and drivers, we recommend seeking assistance from the manufacturer, a computer shop, or a qualified PC technician.

## Minimum system requirements

Make sure your PC's hardware meets the system requirements of any software you are intending to use your Focusrite interface with. Here are some examples:

- [Ableton Live](#)
- [Cubase](#)
- [FL Studio](#)
- [Pro Tools](#)
- [Reaper](#)
- [Reason](#)
- [Studio One](#)

## Hardware recommendations and drivers

### Follow the assembly instructions

When assembling or modifying your PC's hardware and components, follow the manufacturer's recommendations. For example, ensure you install the GPU, RAM, and storage drives into the ports recommended.

### Update system drivers and BIOS

Install or update the relevant drivers and BIOS for your motherboard, CPU, GPU and other components. To do this, you will need to the manufacturer and model of the computer or its components, and then download the relevant drivers and BIOS from their website or using their applications.



We have seen that systems running a BIOS that predates the operating system (e.g., a BIOS version that predates Windows 11) can cause issues. It's particularly important that you update the BIOS if this is the case with your system.

Experience, AMD Software, Creative Cloud, Creative Cloud Desktop App Center etc.).

These apps may be able to update the drivers for you. You can also find many of your Components in **System Information**:

Find your hardware components in System Information

Click  **Start**, type 'System Information' and select **System Information**.

In the **System Information** window that opens, you can find the following in the **System Summary** category:

- **System Manufacturer** – the manufacturer of your PC or motherboard
- **System Model** – the model of your PC or motherboard
- **Processor** – the manufacturer and model of your CPU
- **BIOS Version/Date** – the version and date of the BIOS currently installed on your motherboard
- **BaseBoard Manufacturer** – the manufacturer of your motherboard
- **BaseBoard Version** – the model of your motherboard

Your GPU can be found in System Information under **Components → Display → Name**.

Manufacturers may provide versions of their drivers designed for stability in professional applications, such as music, audio and video production. For example, AMD and NVIDIA provide a PRO and Studio version of their GPU drivers respectively, and these may be more stable.

## Disable 'C-states' in the BIOS

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To save energy when the CPU is idle, the CPU can be commanded to enter a low-power mode. Different processors have varying arrays of power modes collectively called 'C-states' or 'C-modes'. Different processors support different numbers of C-states – some don't support C-states at all.

We've found that having C-states enabled can cause adverse performance on some systems. If you're having audio performance issues, we recommend preventing your processor from entering power-saving C-states.

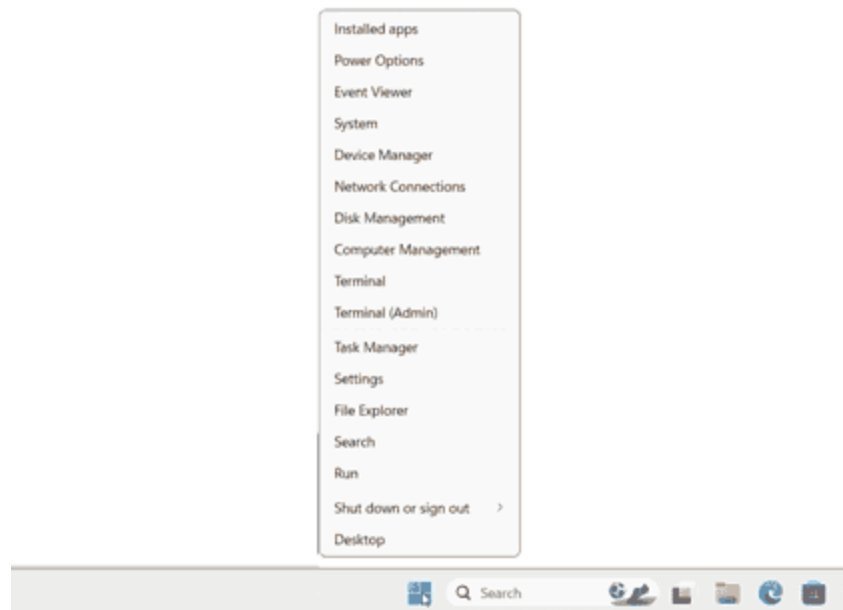
doing this. Focusrite has a dedicated page for this, as the BIOS layout and the available options differ greatly from one machine to another.

## Optimising Windows

### Disable Startup apps

Sometimes your PC will launch apps automatically upon startup. This can negatively affect your computer's performance. To prevent unwanted apps from launching at startup, follow these steps:

1. Right-click  **Start** and open **Task Manager**.
2. Click  **Navigation** in the top left corner and select **Startup apps** (*Windows 10: select the **Startup** tab*).
3. Resize the window if necessary and click **Start-up impact** — this will re-categorise the programs in order of high to low.
4. Right-click on any apps that you do not need at startup and select **Disable**.



We strongly recommend disabling 'Co-pilot' or 'Cortana', as these program can sometimes attempt to use the inputs on your audio interface at inopportune

## Remove bloatware and unwanted apps

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Some computers will be sold with applications installed that aren't needed, or sometimes other apps will install additional software to your system that is not required – often referred to as bloatware. This software may run in the background, interfering with audio or using processing power without your knowledge.

To remove unwanted and unnecessary apps, follow these steps:

1. Click  **Start**, type 'Add or remove programs' and select **Add or Remove Programs** – or click the button below.

Open Add or remove programs

2. Scroll through the list of apps that appear, and find any you do not recognise.
3. Enter the name of the app into your web browser to search for it online, so you can confirm what it does and whether it is required.
4. If you confirm that an app is not needed, click ... **More** and then **Uninstall** (*Windows 10: select the app and click **Uninstall***).
5. Repeat these steps for any apps you do not recognise.

## Third-party antivirus and anti-malware

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Third-party antivirus and anti-malware software (i.e., **NOT** Windows Security or Defender) can interfere with audio interface driver installation and operation. These programs may mistakenly flag parts of the driver as dangerous during installation or cause glitches afterward by altering the timing of system processes.

You may find using Windows Security and Defender is more stable: [Stay protected with Windows Security - Microsoft Support](#).

## Custom power plan

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A power plan is a collection of hardware and system settings that manages how your machine uses and conserves power. Creating a custom power plan will prevent any power settings from interfering with audio recording and playback.



1. Click **Start**, type **power options** and select **power options**.
2. In **Power Options** window that opens, select **Create a power plan** from the options on the left-hand side.
3. Select **High Performance** and enter a **Plan name** — for example, 'Audio Workstation'.
4. Click **Next**.
5. For both, **Turn off the display** and **Put the computer to sleep**, select **Never**.
6. Click **Create**.

## Change the plan settings

You will now be on the **Power Options** page with your custom power plan selected. Customise the power plan with these steps:

7. Click **Change plan settings** next to your new custom power plan
8. Click **Change advanced power settings**.

First, In the **Power Options** window, disable **USB selective suspend** for your custom power plan:

9. Expand **USB settings**, then **USB Selective suspend setting**.

Next, to prevent your computer reducing power to PCIe devices, disable **PCI Express Link State Power Management**:

12. Expand **PCI Express**, then **Link State Power Management**.
13. Next to **Setting**, select **Disable**.
14. If you have a laptop, instead of Setting, two other options will appear: **On battery** and **Plugged in**. Select **Disable** for both.

Finally, prevent your computer reducing power to the processor:

15. Expand **Processor power management**, then **Maximum** and **Minimum processor state**.
16. Next to **Setting**, enter **100** for both.
17. If you have a laptop, instead of Setting, two options will appear for both maximum and minimum: **On battery** and **Plugged in**. Enter **100** for all options.
18. Click **OK**.

## USB power management settings

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This prevents your PC automatically turning off USB ports:

1. Right-click  **Start** and open **Device Manager**.
2. Expand **Universal Serial Bus Controllers**.
3. Right-click the first device and select **Properties**.
4. Go to the **Power management** tab.
5. Untick **Allow the computer to turn off the device to save power**.
6. Apply steps 3 to 5 to each device within **Universal Serial Bus Controllers** – ignore any devices that do not have a Power Management tab.

## Disable Exclusive Mode

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Windows has a feature that allows some applications to take exclusive control of a device, with the primary purpose of lowering latency. However, it can cause problems if that application does not properly allow other applications access to the driver as well. Therefore, we recommend you disable this feature:

1. Right-click  **Start** and open **Settings** – or click the button below.
2. Go to **System** and select **Sound**.

Open Sound Settings

3. Scroll down to **Advanced** and select **More sound settings**  
(*Windows 10: select **Sound control panel***).
4. In the **Sound** window that opens, go to the **Playback** tab.
5. Right-click on your Focusrite audio device and select **Properties**.
6. Go to the **Advanced** tab.
7. Untick **Allow applications to take exclusive control of this device** – the box beneath will untick automatically.

7. Click **OK**.

10. Reboot your computer to apply the changes.

## Disable system sounds

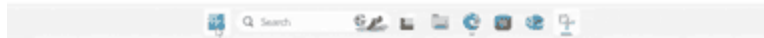
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Sometimes system sounds can interrupt your recordings. Here's how to disable all system sounds:

1. Click  **Start**, type 'Settings' and open **Settings** – or click the button below.
2. Go to **System** and select **Sound**.

Open Sound Settings

3. Scroll down to **Advanced** and select **More sound settings**  
(*Windows 10: select **Sound control panel***).
4. In the **Sound** window that opens, go to the **Sounds** tab.
5. Under the **Sound Scheme**, select **No Sounds**.
6. Click **OK**.



## Set processor scheduling to Background services

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1. Right-click  **Start** and open **System** – or click the button below.

Open System

2. Click **Advanced system settings**.
3. In the **System Properties** window, go to the **Advanced** tab.
4. Under **Performance**, click **Settings....**
5. In the **Performance Options** window, go to the **Advanced** tab and set **Processor scheduling** to **Background Services**.
6. Click **OK**.



## Free up space

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Don't let your computer's boot drive get too full. Always make sure you leave more than 20% of your computer's main hard drive empty. Here's how to check how much storage has been used:

1. Click **Start**, type 'Settings' and open **Settings** – or click the button below.
2. Go to **System**.
3. Select **Storage**.

Open Storage

The bar will show how much storage has been used and is free. You can remove files you no longer need or move them to external drives and cloud storage.





## Disk clean-up

You can also use **Disk clean-up** to remove temporary files and files or folders you no longer need, saving you a lot of space:



The clean-up operation can take a few minutes to up to an hour, so only try this if you have time to spare.

1. Click  **Start**, type 'Disk Clean-up' and open **Disk Clean-up**.
2. Select your **C:** drive.
3. Tick everything under **Files to delete:** and click **OK**.
4. If a box appears asking **Are you sure you want to permanently delete these files?** Click **Delete Files**.



5. Go back to **Disk Clean-up**, select the **C:** drive again, and click **Clean up system files**.
6. Select **C:** drive once more, tick everything again, and click **OK** — this process may take a while, especially if you've never done this before.
7. If a dialogue box appears asking **Are you sure you want to permanently delete these files?**, click **Delete Files**.



Don't shut down or restart your computer during the clean-up operation. The window will disappear once it's finished.

If you find you are experiencing glitchy audio, dropouts, or disconnections after completing all the steps above, please see our glitchy audio troubleshooting guide:

[Troubleshooting glitchy audio on Windows](#)

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